

Cartan Weyl Basis: $\{J^a\}$: Generators of $\mathfrak{g}$
i.e. Basis. $[J^a, J^b] = i f^{ab}_c J^c$

$(J^a)^\dagger = (J^a)$: $\mathbb{R}$ -Hermitian operators

$\mathfrak{g}$: Semi-simple $\Rightarrow \mathfrak{g} = \bigoplus_i \mathfrak{g}_i$

$\mathfrak{g}_i$: simple

$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{g}_i$ $\mathfrak{h}$: Cartan subalg.

• Cartan: Maximal set of commuting Hermitian generators $H_i, i=1, \dots, \dim(\mathfrak{g})$ $[H_i, H_j] = 0$
 $\forall i, j$.

* Now, taking this subgroup $H^i = \sum c_i J^i$

• $\mathfrak{h}$: Cartan, then a root of $\mathfrak{g}$ is given by
 $\alpha \in \mathfrak{h}^*$ if $\exists X \in \mathfrak{g}$ s.t.
 $[H, X] = \alpha(H)X$

$$\text{ad}_H(X) = \alpha(H)X$$

$\leadsto$ Root Space: $\mathfrak{g}_\alpha$ given a root α the root space
 $\mathfrak{g}_\alpha \subset \mathfrak{g}$ is the set $X \in \mathfrak{g}$ s.t.
 $[H, X] = \alpha(H)X$

• Root space $\mathfrak{g}_\alpha$ eigenvectors of ad_H .
 $\mathfrak{g} = \bigoplus_\alpha \mathfrak{g}_\alpha$

• ex: $\mathfrak{g}_0$ $\alpha_0: \mathfrak{h} \rightarrow \mathbb{C}, \alpha_0(X) = 0 \quad \forall X$
 $\Rightarrow [H, X] = \alpha_0(H)X = 0$

$$\mathfrak{g}_0 = \{X \in \mathfrak{g} \mid [H, X] = 0 \quad \forall H \in \mathfrak{h}\}$$

i.e. $\mathfrak{g}_0$: centralizer of $\mathfrak{h}$ and so

$$\boxed{\mathfrak{g}_0 = \mathfrak{h}}$$

$$\boxed{\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha}$$

* Let us take roots of $\mathfrak{h}$:

Here $\rho: \mathfrak{g} \rightarrow V$
 $H: \mathfrak{h} \rightarrow \text{End}(V)$

Let $\{h_i\}_{i=1}^{\dim(\mathfrak{h})}$ span a basis of $\mathfrak{h}$, then
 $H(h_i) \equiv H_i$

H_i : Endomorphism on the vector space V .

• Likewise, we may take the reps of $\mathfrak{g}_\alpha$

$$E: \mathfrak{g}_\alpha \rightarrow \text{End}(V)$$

Note $\mathfrak{g}_\alpha$ are One-dimensional and so they are (up to $\mathbb{Z}_2$) unique.

$$E(X_\alpha) \equiv E^\alpha: V \rightarrow V$$

$$X \in \mathfrak{g}_\alpha$$

$$[H, X] = \alpha(H)X$$

↳ In the above rep:

$$[H_i, E^\alpha] = \alpha(H_i) E^\alpha$$

$$\alpha^i \equiv \alpha(H_i)$$

$$\rho([H, X]) = \rho(\alpha(H)X)$$

$$[\rho(H), \rho(X)] = \alpha(H) \rho(X)$$

$$\alpha = (\alpha^1, \dots, \alpha^{\dim(\mathfrak{g})}) \text{ root}$$

E^α : ladder op.

$$[H_i, E^\alpha] = \alpha^i E^\alpha \text{ take } \dagger$$

$$[(E^\alpha)^\dagger, (H_i)^\dagger] = \alpha^i (E^\alpha)^\dagger$$

$$[H_i, (E^\alpha)^\dagger] = (-\alpha^i) (E^\alpha)^\dagger$$

For which we conclude $(E^\alpha)^\dagger = E^{-\alpha}$ and

$-\alpha$ is a root $-\alpha \in \mathfrak{h}^*$

$$\Delta \equiv \{\text{Set of roots}\}$$

$$\Delta = \Delta^+ \oplus \Delta^-$$

• Roots are eigenvalues of the adjoint rep.

$$\text{ad}: \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$$

Adjoint rep: the underlying vector space is $\mathfrak{g}$ if self!

$$\text{ad}(X): \mathfrak{g} \rightarrow \mathfrak{g}$$

$$\text{ad}(X) \equiv \text{ad}_X$$

$$\text{ad}_X(Y) \equiv [X, Y]$$

• Now, $\text{ad}_{H^i}(E^\alpha) = \alpha^i E^\alpha$

so it is clear that α^i is an eigenvalue

But, since $\mathfrak{g}$ is a vector space then E^α is a vector

→ The vector space we are looking at now is $\text{End}(\mathfrak{g})$: space of linear maps

$$E^\alpha \equiv |\alpha\rangle \in \text{End}(\mathfrak{g})$$

$$H^i \equiv |H^i\rangle \in \text{End}(\mathfrak{g})$$

↙ not sure

$$\text{ad}(H^i) E^\alpha = \alpha^i E^\alpha \quad \mapsto \quad H^i |\alpha\rangle = \alpha^i |\alpha\rangle$$

$$V = \text{End}(\mathfrak{g}), \text{ then } \text{ad}: \mathfrak{g} \rightarrow V$$

$$\text{ad}(H^i) \in V$$

$$\alpha \in \mathfrak{h}^* \quad \alpha: \mathfrak{h} \rightarrow \mathbb{C}$$

$\dim \text{adj}$ is equal to the $\dim$ of the alg.

roots $\alpha \in \mathfrak{h}^*$ has r degeneracy.

i.e. $r: \mathfrak{h} \rightarrow \mathbb{C}$ and each $\mathfrak{g}_\alpha$ is 2-dim.

$$\dim(\text{adjoint}) = \dim(\mathfrak{g})$$

$$* \quad [H^i, [E^\alpha, E^\beta]] = (\alpha^i + \beta^i) [E^\alpha, E^\beta]$$

(from Jacobi)

$$\text{i.e. } [\mathfrak{g}^\alpha, \mathfrak{g}^\beta] = \mathfrak{g}^{\alpha+\beta} \quad \text{if } \alpha + \beta \in \Delta$$

$$[E^\alpha, E^\beta] = 0 \quad \text{if } \alpha + \beta \notin \Delta$$

• If $\alpha = -\beta$, then $[E^\alpha, E^\alpha]$ commutes with all of H^i which is only possible if $[E^\alpha, E^\alpha] = \sum \dot{c}_i H^i$

• If $\alpha = -\beta$, then $[E^\alpha, E^\alpha] = 0$ if $\alpha \notin \Delta$

$$[E^\alpha, E^{-\alpha}] = \sum c_i H^i$$

$$c_i = \frac{2\alpha_i}{|\alpha|^2}$$

$$|\alpha|^2 = \sum_{i=1}^r \alpha_i^2$$

$$\alpha \cdot H = \sum_{i=1}^r \alpha_i H^i$$

$$[E^\alpha, E^{-\alpha}] = \frac{2\alpha \cdot H}{|\alpha|^2}$$

$$[H^i, H^j] = 0$$

$$[H^i, E^\alpha] = \alpha^i E^\alpha$$

$N_{\alpha, \beta}$: Constants

$$[E^\alpha, E^\beta] = \begin{cases} N_{\alpha, \beta} E^{\alpha+\beta} & , \text{ if } \alpha + \beta \in \Delta \\ \frac{2}{|\alpha|^2} \alpha \cdot H & , \text{ if } \alpha = -\beta \\ 0 & , \text{ otherwise} \end{cases}$$

Killing Form: $\tilde{K}(X, Y) = \text{Tr}(\text{ad}(X) \text{ad}(Y))$

$$K(X, Y) = \frac{1}{2g} \text{Tr}(\text{ad}(X) \text{ad}(Y))$$

Standard basis: $\{J^a\}$ defined to be orthonormal

$$K(J^a, J^b) = \delta^{ab}$$

$$K(H^i, H^j) = \delta^{ij}$$

• Use Killing form to lower or raise indices

$$f_{abc} = \sum_d f^{ad}_c [K(J^a, J^b)]^{-1}$$

• Can use cyclic property to show

$$[E^\alpha, E^{-\alpha}] = K(E^\alpha, E^{-\alpha}) \alpha \cdot H$$

$$K(E^\alpha, E^{-\alpha}) = \frac{2}{|\alpha|^2}$$

→ Main role $K: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$

(i) Let $H^i \in \mathfrak{h}$, then $K(H^i, \cdot) \in \mathfrak{h}^*$

$\therefore \forall \gamma \in \mathfrak{h}^* \exists H^\gamma \in \mathfrak{h}$ s.t.

$$\gamma(H^i) = K(H^i, H^\gamma)$$

→ For $\gamma = \alpha$:

$$\alpha(H^i) = K(H^i, H^\alpha)$$

$$H^\alpha = \alpha \cdot H = \sum_i \alpha_i H_i$$

* This defines an inner product on $\mathfrak{h}^* \times \mathfrak{h}^*$

$$(\gamma, \beta) \equiv K(H^\gamma, H^\beta)$$

$$|\alpha|^2 = (\alpha, \alpha)$$

For $(\cdot, \cdot): \mathfrak{h}^* \times \mathfrak{h}^* \rightarrow \mathbb{C}$ on Δ^+

$$(\alpha, \beta) = K(H^\alpha, H^\beta) = K(\alpha \cdot H, \beta \cdot H)$$

$$= \text{Tr}(\text{ad}(\alpha_i H_i) \text{ad}(\beta_j H_j))$$

$$= \alpha_i \beta_j \text{Tr}(\text{ad}(H_i) \text{ad}(H_j))$$

$$= \alpha_i \beta_j K(H_i, H_j) = \alpha_i \beta_j \delta_{ij}$$

$$= \alpha_i \beta_i$$

Thus, (α, β) defines a Euclidean inner product ^{$\dim(\Delta^+)$}

and we can make an isomorphism

$$\alpha \in \mathfrak{h}^* \rightarrow \alpha \in \mathbb{R}^{\dim(\Delta^+)}$$

* Adjoint $(T^a)_{bc} = -if_{abc}$

Now

* Now look at other reps aside from the adjoint rep.

Let $\rho: \mathfrak{g} \rightarrow \text{End}(V)$ be a finite rep of $\mathfrak{g}$.

Then E^λ is $E(\lambda) \in \text{End}(V)$

Now, let $|\lambda\rangle \in V$ s.t. the operator

$$E^\lambda |\lambda\rangle = \lambda |\lambda\rangle$$

• In particular

$$\rho: \mathfrak{h} \rightarrow \text{End}(V), \text{span}\{H_i\} \text{ is } \mathfrak{h} = \mathfrak{h}$$

$$\text{s.t. } \rho(H_i): V \rightarrow V$$

then we can find a basis of V s.t.

$$\rho(H_i) |\lambda\rangle = \lambda^i |\lambda\rangle$$

$$\lambda = (\lambda^1, \dots, \lambda^r) \text{ weight}$$

$$\lambda \in \mathfrak{h}^* \quad \lambda(H_i) = \lambda^i$$

which already has a natural inner product

$$(\cdot, \cdot)$$

* Note $H^i(E^\alpha|\lambda\rangle) = [2H^i, E^\alpha] + E^{-\alpha}H^i|\lambda\rangle$
 $= \alpha^i E^\alpha|\lambda\rangle + \lambda^i E^\alpha|\lambda\rangle$

$$H^i(E^\alpha|\lambda\rangle) = (\alpha^i + \lambda^i)(E^\alpha|\lambda\rangle)$$

thus, for $E^\alpha|\lambda\rangle \neq 0$ it must be that
 $E^\alpha|\lambda\rangle = \omega|\lambda + \alpha\rangle$

* E^α : raises the weight vector by α
 $E^{-\alpha}$: lowers α

$$(E^\alpha)^{p+1}|\lambda\rangle \sim E^\alpha|\lambda + p\alpha\rangle = 0$$

$$(E^{-\alpha})^{q+1}|\lambda\rangle \sim E^{-\alpha}|\lambda - q\alpha\rangle = 0$$

for any root α .

$\Rightarrow \exists$ some state which is in $\ker(E^\alpha)$

Cartan Matrix:

$$A_{ij} \equiv \frac{2(\alpha_j, \alpha_i)}{|\alpha_j|^2}$$

Fundamental weights $\{\omega_i\}$

$\Rightarrow \omega_i$ and α_i live in the same space

Simple roots form a basis of Δ^+

$$\{\alpha_{(i)}\}_{i=1}^r$$

$$\alpha = \alpha^i \alpha_{(i)}$$

$$\alpha = \tilde{\alpha}^i \omega_{(i)}$$

$$(\alpha_{(i)}, \alpha_{(j)}) = |\alpha_{(i)}|^2 \delta_{ij}$$

$$\alpha_{(i)}^v \equiv \frac{2\alpha_{(i)}}{|\alpha_{(i)}|^2}$$

Defined by $(\omega_{(i)}, \alpha_{(j)}^v) = \delta_{ij}$

* Take a weight $\lambda = (\lambda_1, \dots, \lambda_r)$ in the basis $\{\omega_{(i)}\}$

$$\lambda = \sum_{i=1}^r \lambda_i \omega_{(i)}$$

λ_i : Dynkin label

$$\lambda_i = (\lambda, \alpha_{(i)}^v)$$

* Thus, for any weight $\lambda = (\lambda_1, \dots, \lambda_r)$ are the Dynkin labels

We now relate the simple roots Δ^+ to the fundamental weights

$$A_{(i)(j)} = \frac{2(\alpha_{(i)}, \alpha_{(j)})}{|\alpha_{(i)}|^2}$$

$$\alpha_{(i)} = \sum_j \beta^j \omega_{(j)} \Rightarrow (\alpha_{(i)}, \alpha_{(j)}^\vee) = \sum_j \beta^j (\omega_{(j)}, \alpha_{(j)}^\vee)$$

$$\beta^j = A_{(j)(i)}$$

$$\therefore \alpha_{(i)} = \sum_j A_{ij} \omega_{(j)}$$

* The Dynkin labels of a simple root is the elements of the Cartan Matrix: A_{ij} *

→ i^{th} row of A are the Dynkin labels for the simple root $\alpha_{(i)}$

$$A_{ij} = \begin{pmatrix} \alpha_1^\top \\ \vdots \\ \alpha_r^\top \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{matrix} \leftarrow \alpha_1 \\ \leftarrow \alpha_2 \end{matrix}$$

$$\alpha_1 = 2\omega_1 + 1\omega_2$$

$$\alpha_2 = \omega_1 + 2\omega_2$$

$$\text{ex: } \alpha_{(i)} = \sum_j A_{ij} \omega_{(j)} = \begin{pmatrix} A_{i1} \\ A_{i2} \\ \vdots \\ A_{ir} \end{pmatrix}$$

* For a highest weight Θ , $E^{-\alpha}|\Theta\rangle = n|\Theta - \alpha\rangle$
until $(E^{-\alpha})^n |\Theta\rangle = 0$

$$\text{SU}(2): A = (2), \alpha_1 = 2\omega_1$$

$$\otimes \quad \otimes \quad \otimes \quad \otimes \quad \otimes \quad \otimes$$

$$-\alpha_1 - \omega_1 \quad 0 \quad \omega_1 \quad \alpha_1 \quad 3\omega_1$$

* Highest weight state

$$\lambda = [\lambda^1, \dots, \lambda^r]$$

Adjoint rep: Highest root $\Theta = \alpha_1 + \alpha_2 + \dots + \alpha_r$

ex: $\mathfrak{sl}(3)$: $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ A_2


$$\alpha_1 = 2\omega_1 - \omega_2 = (2, -1)$$

$$\alpha_2 = -\omega_1 + 2\omega_2 = (-1, 2)$$

→ Take the highest weight

$$[1, 1] = \lambda = \omega_{(1)} + \omega_{(2)}$$

Note this is the adjoint rep

$$\alpha_1 + \alpha_2 = \omega_1 + \omega_2$$

$$[1, 1]$$

$$\begin{array}{cc} -\alpha_1 \swarrow & \searrow -\alpha_2 \end{array}$$

$$\begin{array}{cc} [-1, 2] & [2, -1] \end{array}$$

$$\begin{array}{cc} -\alpha_2 \swarrow & \searrow -\alpha_1 \end{array}$$

$$[0, 0]$$

$$\begin{array}{cc} -\alpha_1 \swarrow & \searrow -\alpha_2 \end{array}$$

$$\begin{array}{cc} [-2, 1] & [1, -2] \end{array}$$

$$\begin{array}{cc} -\alpha_2 \swarrow & \searrow -\alpha_1 \end{array}$$

$$[-1, -1]$$